Bhabishya Shrestha

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Education

The University of Texas at Austin — Austin, TX
Master of Science in Artificial Intelligence — Aug 2026 – Present (Expected May 2028)

The University of Texas at Dallas — Richardson, TX
Master of Science in Computer Engineering (Fast Track), GPA: 3.53 — May 2025
Bachelor of Science in Computer Engineering (Cum Laude), GPA: 3.78 — May 2024

Relevant Coursework: Ethics in AI, Natural Language Processing, Machine Learning, Artificial Intelligence, Linear Algebra, Probability & Statistics, Data Structures, Operating Systems.

Skills

AI & Machine Learning: LLMs, RAG Architectures, Agentic Frameworks (MCP), OpenCV, Tesseract, Time-Series Forecasting
Languages & Frameworks: Python, C++, Java, JavaScript/TypeScript, SQL, GraphQL, React, React Native, Next.js, FastAPI, Kotlin
Cloud & DevOps: AWS (Lambda, DynamoDB, S3, Cognito), Azure, Docker, Kubernetes, Jenkins, CI/CD, Grafana
Databases: MSSQL, PostgreSQL, Oracle, MySQL, MongoDB, Db2, Cassandra, DynamoDB; Database Administration (SSMS)
Certifications: AWS Cloud Practitioner, Agile Foundations (AFC-AICI)

Work Experience

Site Reliability Engineer (SRE) — Visa, Austin, TX — Oct 2025 – Present

- **GenAI Initiative Lead:** Architecting an agentic AI framework utilizing the **Model Context Protocol (MCP)** to automate database triage for large-scale incidents across MSSQL, PostgreSQL, Oracle, MySQL, MongoDB, Db2, and Cassandra, integrated with Grafana and centralized logging (ClickHouse).
- **Incident Automation:** Built automation that combines the triage tooling with historical ServiceNow incident data to surface similar past incidents and SOP-based resolution guides on every incident.
- **Applied ML:** Exploring **time-series forecasting** models to support infrastructure capacity planning (early stage).
- **Database Reliability:** Managing high-availability MSSQL clusters, performing critical patching, replication management, and access control governance via SSMS for global distributed payment systems.
- **Automation:** Engineering Python and Shell automation scripts to streamline database reliability audits and smoke tests, ensuring 99.999% system availability for critical payment rails.

Solutions Engineer — Allegion US, Austin, TX — May 2025 – Oct 2025

- Served as the technical SME for mobile credential platforms, onboarding B2B customers on REST APIs and SDK integrations.
- Conducted deep-dive root cause analysis for escalated issues across distributed cloud and mobile platforms, collaborating with engineering to define permanent fixes.
- Refined API documentation based on developer feedback to streamline partner integrations.

Mobile Engineer Intern — Accessibility Team — Fidelity Investments — Jun 2024 – Aug 2024

- Produced reusable Jetpack Compose components and automated accessibility lint rules, improving UI consistency and secure-coding standards across an internal Android app.
- Delivered a technical pitch to the VP of Engineering, advocating for the enterprise-wide adoption of Jetpack Compose for improved security and development efficiency.
- Drove Agile story refinement with product and QA teams; led peer reviews emphasizing performance and WCAG compliance.

Enterprise Technology Platform Intern — Southern Glazer's Wine & Spirits — May 2023 – Aug 2023

- Strengthened CI/CD pipelines using Jenkins and JUnit to automate testing, increasing code coverage from 65% to 88%.
- Presented research findings on **sentiment analysis** to leadership, focusing on data privacy, secure data handling practices, and customer feedback loops to optimize product distribution.
- Performed root cause analysis for REST API defects in production Java microservices.

Co-Founder & Full Stack Developer — Hidden Gems (Startup) — Aug 2022 – Mar 2024

- Architected a cloud-native marketplace and React Native (Expo) mobile app connecting local restaurants with corporate catering clients, managing the full SDLC.
- Designed GraphQL schema and REST endpoints, implemented order management and group-order flows, integrated Square for menus and ordering, and enforced role-based access with AWS Cognito.

- Engineered and maintained a resilient AWS backend (DynamoDB/Lambda); built CloudWatch dashboards to monitor system health.

Projects

Breast Cancer Research RAG Chatbot qoltbot.netlify.app

- Integrated OpenAI-based LLMs with FastAPI to offer data-backed medical responses using **RAG architecture**.
- Built a secure React front end connected to AWS S3 for knowledge base and Cognito for authentication.

Motherboard Identifier **1st Place, UT Design Expo** tinyurl.com/motherboard-inspector

- Engineered an inspection tool in Python using **OpenCV and Tesseract**, achieving 100% text-recognition accuracy.
- Packaged pipeline into a cross-platform CLI, enabling technicians to batch-process images locally on Windows and Linux.

Aura Finance aura-finance-tool.vercel.app

- Architected a local-first personal finance dashboard using React, Zustand, Dexie.js (IndexedDB), and Firebase, deployed via Vercel CI/CD.
- Implemented Firebase Auth with per-user Firestore security rules for data isolation, AI-assisted statement import (CSV, PDF, images), and automated testing with Vitest.

Leadership

Vice President, Tau Beta Pi (Xi Chapter) Spring 2022 – Spring 2024

- Selected as among the top eighth of the junior year class and major department (ECS) for acceptance into the organization.
- Led a board of 5 officers to organize and execute professional development events, managing the chapter budget and boosting engagement among 40+ active members.

Technical Director, IEEE at UTD Spring 2022 – Spring 2023

- Established the organizational structure for "The Forge," a technical incubator fostering a culture of innovation.
- Supervised a team of 4 Project Managers, providing technical direction and guidance to ensure timely and high-quality program delivery.
- Facilitated communication among program managers and faculty advisors to align project goals with the IEEE mission.

Project Manager, IEEE at UTD Spring 2021 – Spring 2022

- **Facial Security System (IoT):** Led a team of 5 engineering students to design a micro-controller IoT security system. Oversaw the SDLC utilizing the Magic Mirror library and **OpenCV** for real-time facial detection.
- **Arduino Robotic Arm:** Directed the design and assembly of a 4-axis robotic arm using C++ and Arduino Mega 2650. Engineered control logic for the HC-05 Bluetooth Module enabling remote control via Android Joystick.